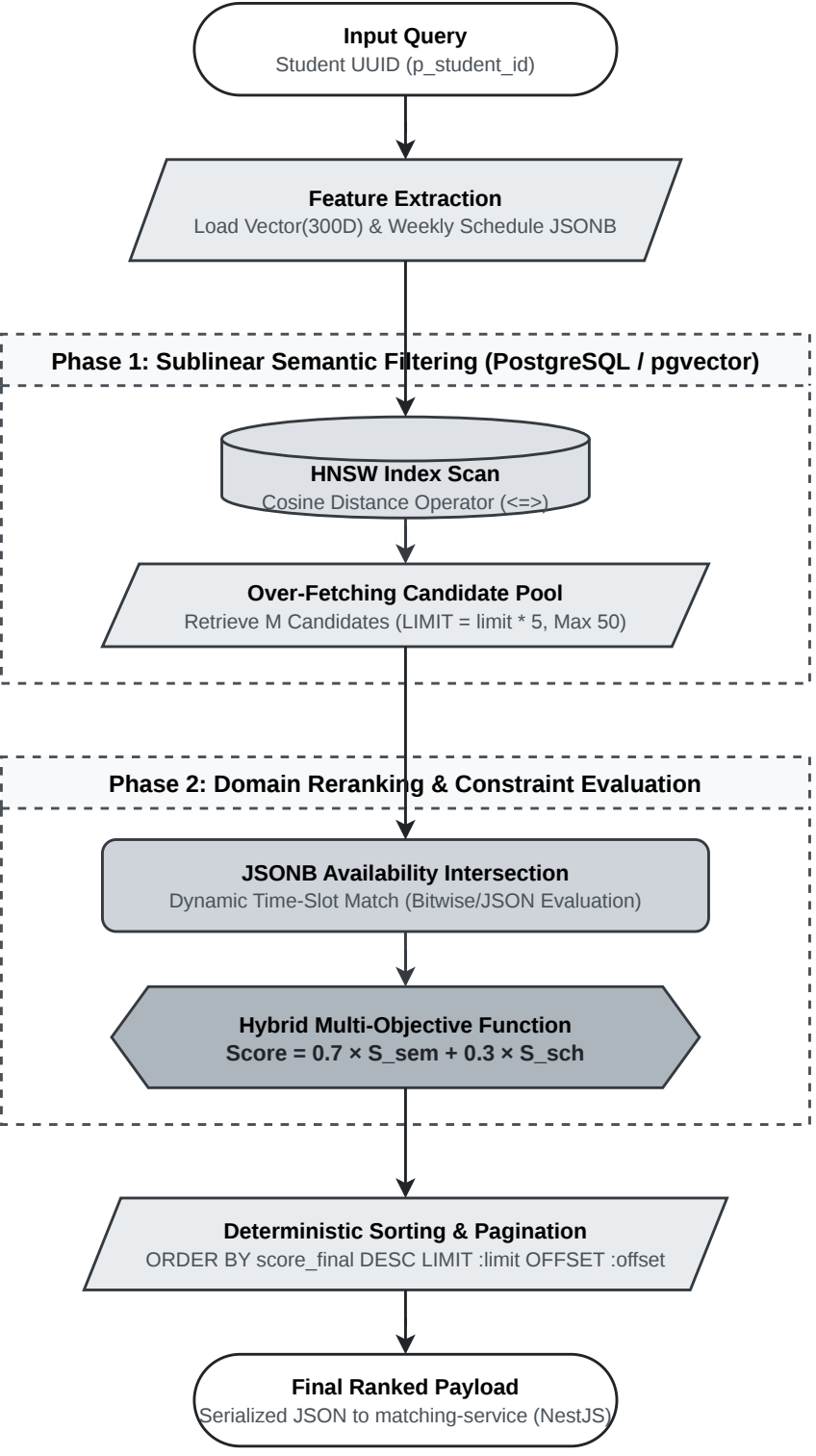
Input Query
Student UUID (p_student_id)
Feature Extraction
Load Vector(300D) & Weekly Schedule JSONB
Phase 1: Sublinear Semantic Filtering (PostgreSQL / pgvector)
HNSW Index Scan
Cosine Distance Operator (<=>)
Over-Fetching Candidate Pool
Retrieve M Candidates (LIMIT = limit * 5, Max 50)
Phase 2: Domain Reranking & Constraint Evaluation
JSONB Availability Intersection
Dynamic Time-Slot Match (Bitwise/JSON Evaluation)
Hybrid Multi-Objective Function
Score = 0.7 × S_sem + 0.3 × S_sch
Deterministic Sorting & Pagination
ORDER BY score_final DESC LIMIT :limit OFFSET :offset
Final Ranked Payload
Serialized JSON to matching-service (NestJS)